

3.6 Alternating Series

So, far we have been focusing on series which ultimately have all positive terms. In this section we will consider a special kind of series that has negative terms.

Definition: An alternating series is

Example:

Note:

Example 1: Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

- a) Using the notation from class what does a_n equal in this case?
- b) TRUE or FALSE: $a_{n+1} \leq a_n$ for all n and $\lim_{n \rightarrow \infty} a_n = 0$.
- c) Compute the first, second, third, fourth, fifth, and sixth partial sum of this series. We see that:

- d) TRUE or FALSE: The even partial sums for this series form an increasing sequence.
- e) TRUE or FALSE: The sequence of even partial sums for this series is bounded above by 1.
- f) Do you think this series is convergent or divergent? Explain.

Note: This example uses many of the ideas in the proof of the Alternating Series Test.

Theorem (Alternating Series Test): If the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ (where $a_n > 0$) satisfies the following two conditions:

Proof: See the textbook.

Example 2: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1} (n+3)}{n(n+1)} \right)$ converges or diverges.

Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^n (3n-1)}{2n+1} \right)$ converges or diverges.

Theorem:

We will omit the proof of this theorem.

Example 4: Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

- a) Find the seventh partial sum of this series.

- b) We know that this series converges. What is an upper bound on how far off the number that you calculated in (a) is from the actual sum of the series.

- c) What is the first partial sum of this series that we can guarantee is within one thousandth of the actual sum (using the theorem)?

Note: Later on in this course we will be able to prove that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln(2)$.

Definition:

Example 5: State (with justification) whether each of the following series is absolutely convergent, conditionally convergent, or divergent.

a)
$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

b)
$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$$

Theorem: If a series $\sum a_n$ is absolutely convergent, then it is convergent.

Proof:

Example 6: Show that the following series converges: $1 - \frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} - \frac{1}{2^5} - \frac{1}{2^6} + \dots$